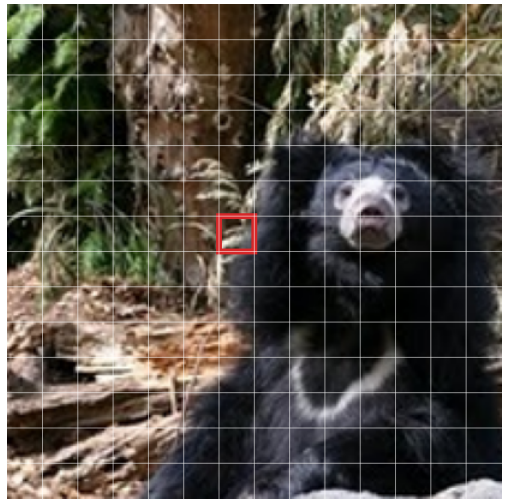
Patch embedding is a local projection and a sparse global linear map

Native patch projection

Input image



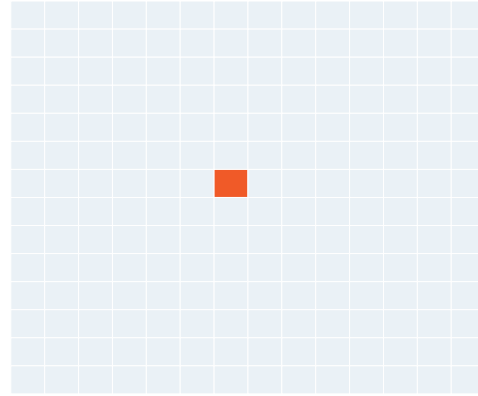
Patch row
$w_c op$

$z_{t,c}$

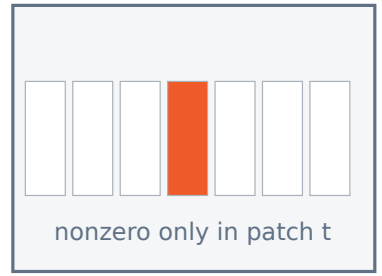
$$z_{t,c} = w_c^\top x_t + b_c$$

Sparse linear form

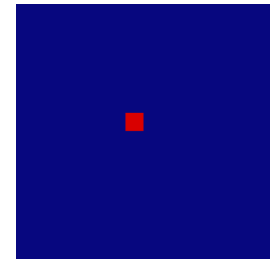
Image coordinates



Sparse row $W_{t,c}$



Learning-image support



$$E_{t,c} = x \odot W_{t,c}$$

Bias changes the scalar value but has no input-coordinate term.